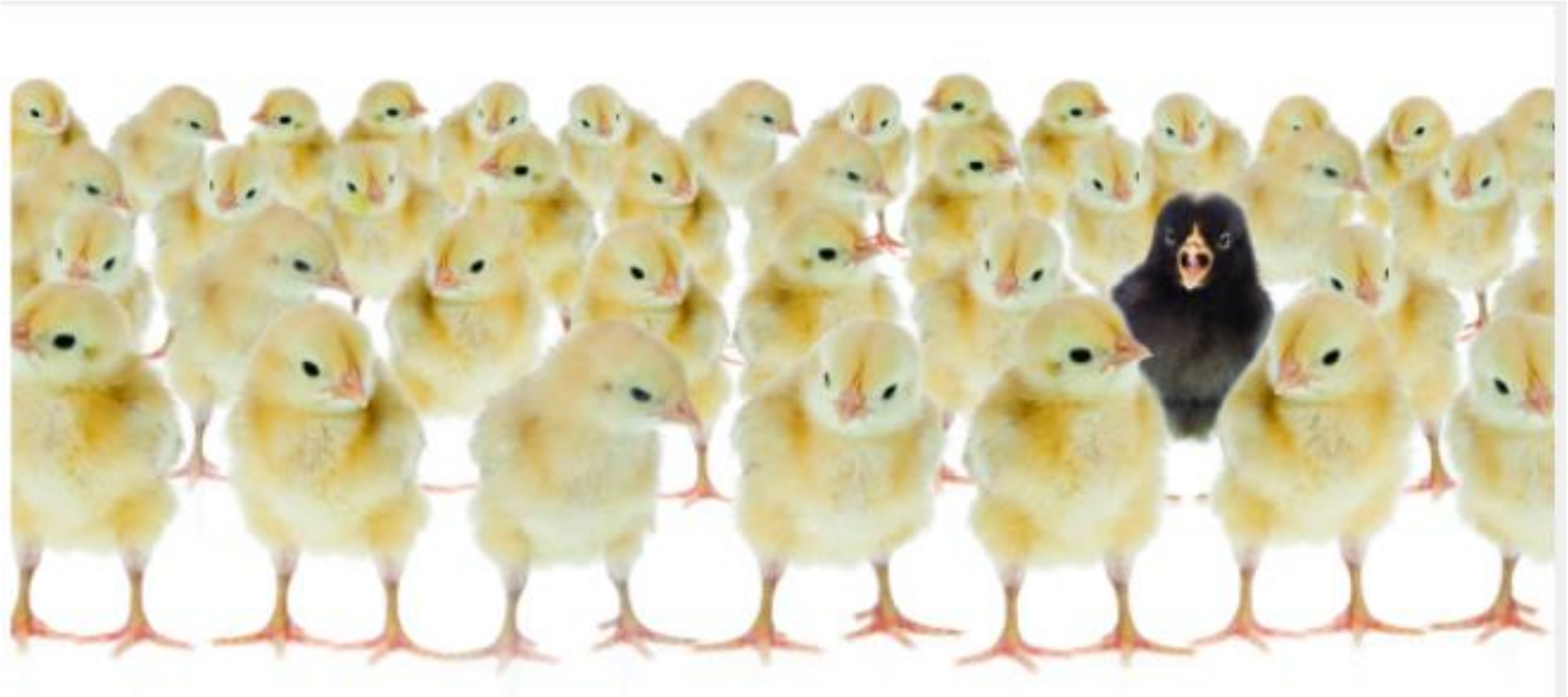

بسم الله الرحمن الرحيم

اَللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَّافِعًا،

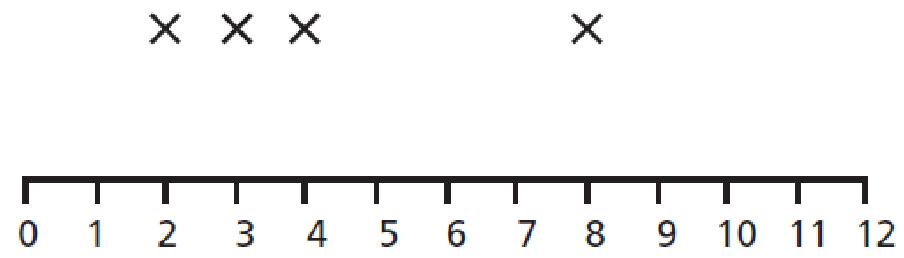
وَرِزْقًا طَيِّبًا، وَعَمَلًا مُّتَقَبَّلًا،

Measures of Dispersion

Week 2



Group A:



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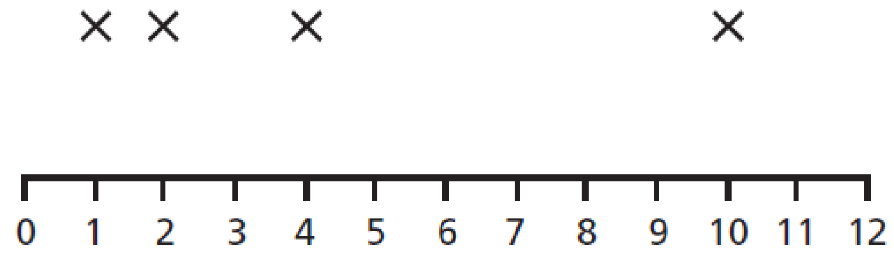
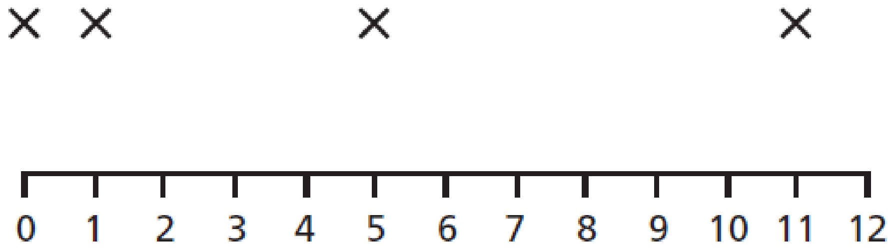
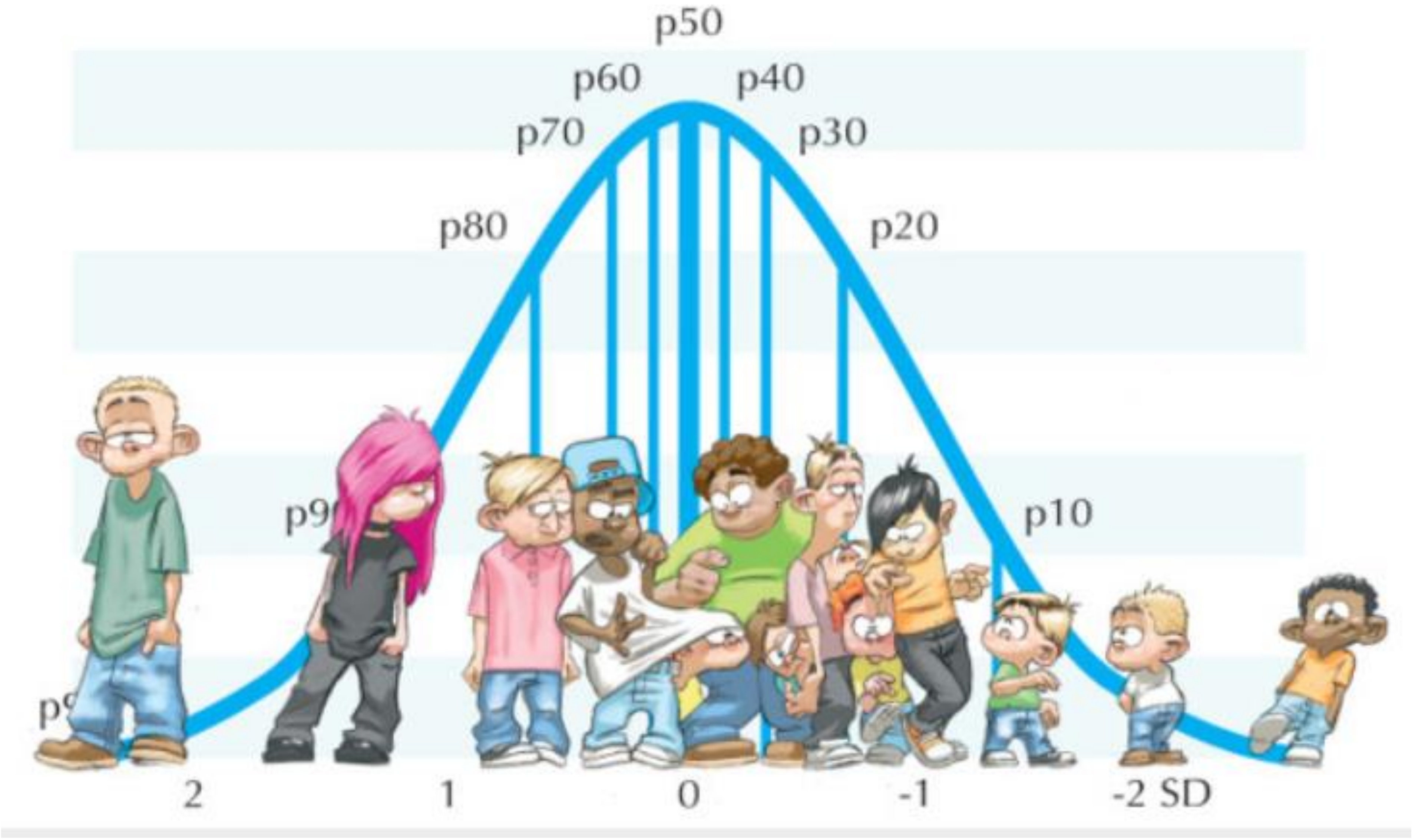


Group C:



Variability



OBJECTIVES OF MEASURING DISPERSION

- To determine the reliability of an average
- To compare the variability of two or more series
- For facilitating the use of other statistical measures
- Basis of Statistical Quality Control

Properties of a Good Measure of Dispersion

- Easy to understand
- Simple to calculate
- Uniquely defined
- Based on all observations
- Not affected by extreme observations
- Capable of further algebraic treatment

Measures of Dispersion

Absolute

- Expressed in the same units in which data is expressed
- Ex: Rupees, Kgs, Ltr, Km etc.

Relative

- In the form of ratio or percentage, so is independent of units
- It is also called **Coefficient of Dispersion**

Measures of Variability

- Range
- Variance
- Standard deviation

The Range

- The *range* is defined as the difference between the largest score in the set of data and the smallest score in the set of data, $X_L - X_S$

- What is the range of the following data:
4 8 1 6 6 2 9 3 6 9

- The largest score (X_L) is 9; the smallest score (X_S) is 1; the range is $X_L - X_S = 9 - 1 = 8$

RANGE (R)

- It is the simplest measures of dispersion
- It is defined as the difference between the largest and smallest values in the series

$$R = L - S$$

R = Range, L = Largest Value, S = Smallest Value

- Coefficient of Range = $\frac{L-S}{L+S}$

PRACTICE PROBLEMS – RANGE

Q1: Find the range & Coefficient of Range for the following data: 20, 35, 25, 30, 15

Ans: 20, 0.4

Q2: Find the range & Coefficient of Range:

X	10	20	30	40	50	60	70
F	15	18	25	30	16	10	9

Ans: 60, 0.75

Q3: Find the range & Coefficient of Range:

Size	5-10	10-15	15-20	20-25	25-30
F	4	9	15	30	40

Ans: 25, 5/7

Range

MERITS

- Simple to understand
- Easy to calculate
- Widely used in statistical quality control

DEMERITS

- Can't be calculated in open ended distributions
- Not based on all the observations
- Affected by sampling fluctuations
- Affected by extreme values

STANDARD DEVIATION

- The concept of standard deviation was first introduced by Karl Pearson in 1893. The standard deviation is the most useful and the most popular measure of dispersion. Just as the arithmetic mean is the most of all the averages, the standard deviation is the best of all measures of dispersion.

Standard Deviation

- Most important & widely used measure of dispersion
- First used by Karl Pearson in 1893
- Also called root mean square deviations
- It is defined as the square root of the arithmetic mean of the squares of the deviation of the values taken from the mean
- Denoted by σ (sigma)
- $\sigma = \sqrt{\frac{\Sigma(X-\bar{X})^2}{N}}$ or $\sqrt{\frac{\Sigma x^2}{N}}$ where $x = X - \bar{X}$
- Coefficient of S.D. $= \frac{\sigma}{\bar{X}}$

Standard Deviation – Individual Series Method Based on Use of Actual Data

- $\sigma = \sqrt{\frac{\Sigma X^2}{N} - \left(\frac{\Sigma X}{N}\right)^2}$

Q3: Calculate the SD of the following data:

16, 20, 18, 19, 20, 20, 28, 17, 22, 20

Ans: 3.13

Variance

The **sample variance**, denoted by s^2, is given by

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}.$$

The **sample standard deviation**, denoted by s, is the positive square root of s^2, that is,

$$s = \sqrt{s^2}.$$

Example

An engineer is interested in testing the “bias” in a pH meter. Data are collected on the meter by measuring the pH of a neutral substance (pH = 7.0). A sample of size 10 is taken, with results given by

7.07 7.00 7.10 6.97 7.00 7.03 7.01 7.01 6.98 7.08.

The sample mean $\bar{x}$ is given by

$$\bar{x} = \frac{7.07 + 7.00 + 7.10 + \cdots + 7.08}{10} = 7.0250.$$

The sample variance s^2 is given by

$$s^2 = \frac{1}{9}[(7.07 - 7.025)^2 + (7.00 - 7.025)^2 + (7.10 - 7.025)^2 + \cdots + (7.08 - 7.025)^2] = 0.001939.$$

Combined Standard Deviation

- It is the combined standard deviation of two or more groups as in case of combined arithmetic mean
- It is denoted by $\sigma_{12} = \sqrt{\dfrac{N_1\sigma_1^{\,2}+N_2\sigma_2^{\,2}+N_1d_1^{\,2}+N_2d_2^{\,2}}{N_1+N_2}}$

where σ_{12} = Combined SD

σ_1 = SD of first group

σ_2 = SD of second group

$d_1 = \overline{X_1} - \overline{X_{12}}$

$d_2 = \overline{X_2} - \overline{X_{12}}$

$$\text{combined mean } \overline{x_{12}} = \frac{n_1\overline{x_1} + n_2\overline{x_2}}{n_1 + n_2} \qquad \text{(if two data sets)}$$

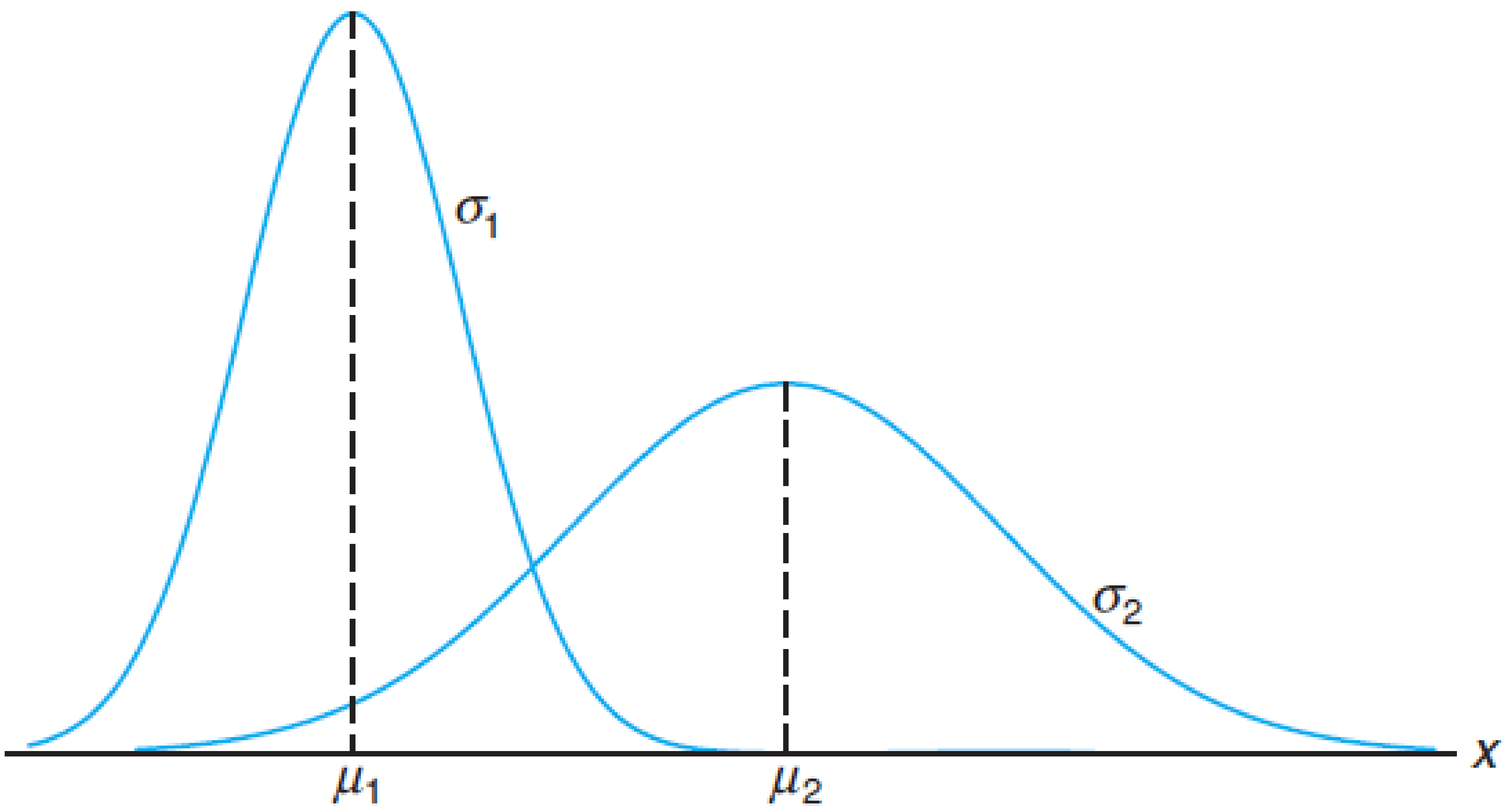
$$\overline{x_{123}} = \frac{n_1\overline{x_1} + n_2\overline{x_2} + n_3\overline{x_3}}{n_1 + n_2 + n_3} \qquad \text{(if three data sets)}$$

Combined standard deviation

$$\sigma_{12} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

$$d_1 = \overline{x_{12}} - \overline{x_1}$$

$$d_2 = \overline{x_{12}} - \overline{x_2}$$


σ1
σ2
μ1
μ2
X

Practice Problems

Q9: Two samples of sizes 100 & 150 respectively have means 50 & 60 and SD 5 & 6. Find the Combined Mean & Combined Standard Deviation.

Ans: 56, 7.46

Important Practice Problems

Q10: The mean weight of 150 students is 60 kg. The mean weight of boys is 70 kg with SD of 10 kg. The mean weight of girls is 55 kg with SD of 15 kg. Find the number of boys & girls and their combined standard deviation.

Ans: 50, 100, 15.28

Q11: Find the missing information from the following:

	Group I	Group II	Group III	Combined
Number	50	?	90	200
SD	6	7	?	7.746
Mean	113	?	115	116

Ans: 60, 120, 8

Coefficient of Variation (C.V.)

- It was developed by Karl Pearson.
- It is an important relative measure of dispersion.
- It is used in comparing the variability, homogeneity, stability, uniformity & consistency of two or more series.
- Higher the CV, lesser the consistency.
- C.V. $= \frac{\sigma}{\bar{X}} x\ 100$

Why Standard deviation?

1.17 A study of the effects of smoking on sleep patterns is conducted. The measure observed is the time, in minutes, that it takes to fall asleep. These data are obtained:

Smokers:	69.3	56.0	22.1	47.6
	53.2	48.1	52.7	34.4
	60.2	43.8	23.2	13.8
Nonsmokers:	28.6	25.1	26.4	34.9
	29.8	28.4	38.5	30.2
	30.6	31.8	41.6	21.1
	36.0	37.9	13.9	

(a) Find the sample mean for each group.

(b) Find the sample standard deviation for each group.

(c) Make a dot plot of the data sets A and B on the same line.

(d) Comment on what kind of impact smoking appears to have on the time required to fall asleep.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